

Power Up **ANSWER KEY**

Early finishers · Grade 7 Accelerated · Unit 2 review

Name _____

1. Words to know

In 2^5 the 2 is the **base** and the 5 is the **exponent**.

2. Squares and cubes (from Unit 2)

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

n	1	2	3	4	5
n^3	1	8	27	64	125

3. Powers of 2 (each one is double the one before)

n	1	2	3	4	5	6	7	8	9	10
2^n	2	4	8	16	32	64	128	256	512	1024

4. Roots and real numbers (use your squares and cubes tables)

a) Each root is between which two whole numbers?

$\sqrt{20}$: 4 and 5

$\sqrt{50}$: 7 and 8

$\sqrt{90}$: 9 and 10

$\sqrt{150}$: 12 and 13

$\sqrt[3]{30}$: 3 and 4

$\sqrt[3]{100}$: 4 and 5

b) Rational or irrational? Circle one.

$\sqrt{49}$: rational (= 7)

$\sqrt{50}$: irrational

$\sqrt[3]{(-27)}$: rational (= -3)

0.333...: rational (= 1/3)

π : irrational

$\sqrt{121}$: rational (= 11)

c) Order from least to greatest: $\sqrt{10}$, 3.2, π , $10/3$

$\pi < \sqrt{10} < 3.2 < 10/3$ ($\pi \approx 3.1416$, $\sqrt{10} \approx 3.1623$, $3.2 \approx 3.2000$, $10/3 \approx 3.3333$)

5. Which is larger? Circle the larger one, or write = if they are equal.

a) 2^5 or 5^2

2^5 (32 vs 25)

b) 3^4 or 4^3

3^4 (81 vs 64)

c) 2^4 or 4^2

equal: both 16

d) 2^{10} or 10^3

2^{10} (1024 vs 1000)

e) 1^9 or 9^1

9^1 (9 vs 1)

Challenge

Find a power between 500 and 600, using an exponent greater than 1. There are four of them. Can you find them all?

$2^9 = 512$, $8^3 = 512$, $23^2 = 529$, $24^2 = 576$